

Mycelic: A Hierarchical Lineage Fabric for Persistent Agent Collectives

Local memory, organizational abstraction, durable transport, and reconstructable collective knowledge

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Abstract. Persistent AI agents increasingly require knowledge that outlives a context window and crosses organizational boundaries without forcing all raw memory into one canonical store. We introduce **Mycelic**, a local-first architecture in which each agent retains evidence-bearing memory inside an independently owned **NeuralGraph**, policy-approved artifacts move through **NATS JetStream**, and organizational knowledge is coordinated through a **3D Lineage Fabric**. The fabric has three typed relation families: horizontal peer coordination at a common abstraction level; vertical semantic abstraction from agent to team, department, subsidiary, and company; and lineage threads that preserve exact derivation paths for drill-down, audit, and failure analysis. Cross-functional scopes such as projects and missions remain overlapping hyperedges rather than being forced into the organization tree. **Tesseract** is the four-face query router over single-hop, multi-hop, temporal, and open-domain retrieval, while a factorized set-valued routing layer fuses semantic, scope, time, lineage, policy, and cost signals to choose a sufficient coalition of agents. The central design principle is *capability coherence without state convergence*: organizational knowledge can become more abstract at higher levels while remaining reconstructable from locally owned evidence. We formalize this topology, define failure-domain survivability over authorized derivations, and specify matched benchmarks against centralized, centralized-graph, replicated-central, federated, flat-distributed, and hierarchical baselines. All headline scores are reserved for controlled measurements under identical models, evidence, policies, and budgets rather than imported from incomparable papers.

Core claim to test. Persistent organizational knowledge need not converge into one global memory. A population of locally owned memory graphs can preserve collective capability if the system maintains enough *authorized, semantically sufficient, and lineage-independent reconstruction paths* and can route queries to them efficiently.

1 Introduction

Long-lived agents create a systems problem that is easy to misname. A conversation transcript is not the same thing as a memory; a memory is not the same thing as knowledge; and a durable message stream is not a knowledge base. When these concepts are collapsed, the resulting architecture becomes difficult to reason about, difficult to secure, and nearly impossible to evaluate causally.

Mycelic separates the stack into four responsibilities. **NeuralGraph** stores local, persistent, evidence-bearing memory for one agent or trust boundary. **JetStream** provides durable event transport and replay between nodes, but is not the canonical semantic store. The **Mycelic Fabric** maintains a sparse global topology of policy-approved knowledge artifacts and capabilities. **Tesseract** is only the query router: it projects a request onto one of four retrieval faces and selects which regions or agents should be consulted.

This separation supports a stronger scientific question than “better memory retrieval”: *when does a distributed population retain a capability after parts of its evidence, routing, permissions, or agents disappear?* The architecture is useful only if it enables experiments that distinguish ordinary replication from genuinely independent reconstructability.

1.1 Contributions

This paper makes five concrete contributions as a research design:

1. A precise separation of **memory**, **knowledge**, **transport**, and **routing** in a local-first multi-agent architecture.
2. A formal **3D Lineage Fabric** for organizational knowledge: horizontal peer coordination, vertical semantic abstraction, and lineage-preserving derivation across levels, with projects and missions represented as cross-cutting scopes.
3. A mathematically explicit **factorized routing topology** that assigns semantics, scope, time, and lineage to representations appropriate to their structure rather than forcing them into a single metric space.
4. A **capability-survivability formalism** over authorized derivation paths and failure domains, including a proposed active-lineage-diversification controller.
5. A matched evaluation protocol that scores centralized, centralized-graph, replicated-central, federated, flat-distributed, hierarchical, and full Mycelic architectures on capability, cost, privacy exposure, and failure survival.

2 System Model: Memory Is Local, Knowledge Is Distributed

Let agent i own a local memory graph

$$G_i = (V_i, E_i, \psi_i), \quad (1)$$

where V_i is a set of persistent memory objects, E_i contains typed local relations, and ψ_i contains local metadata and policy state. There are **no raw cross-NeuralGraph edges**. Cross-node exchange occurs through typed artifacts.

2.1 Memory versus knowledge

A local memory $m \in V_i$ is an evidence-bearing state object: an observed event, extracted fact, decision, episode, artifact pointer, or derived local record with provenance. By contrast, a knowledge claim k is a proposition whose truth for a task is supported by one or more authorized derivations over memories:

$$\mathcal{D}(k) = \{d_1, \dots, d_r\}, \quad d_j : M_j \Rightarrow k. \quad (2)$$

Knowledge therefore need not be stored as a literal sentence. A collective may know k if it can reconstruct k from distributed evidence when asked.

A materialized **KnowledgeArtifact** is an optimization and policy boundary, not a synonym for raw memory. Conceptually,

$$a = (id, owner, z, \Sigma, \tau, \lambda, p, refs), \quad (3)$$

where z is semantic state, Σ is scope, τ is temporal validity, λ is lineage, p is policy/authorization metadata, and $refs$ are opaque pointers back to local evidence.

2.2 Ingestion and local persistence

A foreground agent should not write durable memories directly from each utterance. The local path is

$$\begin{aligned} \text{events} &\rightarrow \text{buffer} \rightarrow \text{normalize} \rightarrow \text{extract} \\ &\rightarrow \text{commit gate} \rightarrow G_i. \end{aligned} \quad (4)$$

The session buffer resolves coreference, temporal phrases, entity aliases, and unfinished facts. Normalization preserves raw text immutably and creates a separate corrected representation. The extractor proposes *candidate knowledge*; a deterministic commit gate decides whether it is new, duplicate, contradictory, unresolved, superseding, or inadmissible.

3 The 3D Lineage Fabric: Organizational Knowledge Without a Global Raw-Memory Graph

The central architectural object is a *multilayer directed knowledge graph*, not a Euclidean coordinate system. “3D”

is shorthand for three orthogonal relation families that answer different questions:

- **Horizontal peer plane (E_H):** who at the same abstraction level can contribute relevant knowledge? Agent-to-agent, team-to-team, and department-to-department routes live here.
- **Vertical abstraction axis (E_V):** what lower-level evidence deserves to become a higher-level organizational representation? Agent memories can yield team artifacts; team artifacts can yield department abstractions; higher levels contain new semantic state rather than raw replicas.
- **Lineage threads (E_L):** exactly which lower-level evidence and transformations support a higher-level artifact? These derivation edges make abstraction reversible enough for audit, drill-down, counterfactual failure, and repair.

Formally, let

$$\mathcal{M} = (V, E_H, E_V, E_L, \ell, \Sigma, \tau, \pi), \quad (5)$$

where $\ell(v)$ is the organizational abstraction level, $\Sigma(v)$ is the set of overlapping scopes attached to v (team, department, project, mission, incident, customer, etc.), $\tau(v)$ is temporal validity, and $\pi(v)$ is policy state. E_L forms a derivation DAG even when E_H and E_V contain richer routing structure.

3.1 Why the organizational hierarchy exists

The hierarchy is not merely a permissions tree. Its purpose is to let representation resolution change with organizational scale. A user’s raw observation may be useful locally; a team may need a repeated operational pattern; a department may need a systemic risk; the company may need a strategic constraint. If A_ℓ is an abstraction operator from level ℓ to $\ell + 1$, then

$$K_{\ell+1} = A_\ell(K_\ell^{(1)}, \dots, K_\ell^{(n)}), \quad (6)$$

with lineage retaining the parent set and derivation operator. Higher-level artifacts must therefore contain *new semantic payload*, not concatenated lower-level memories. A hierarchical query can first operate at the level appropriate to its scope and drill down only when confidence, conflict, or evidence requirements demand it. A simple expected-cost model is

$$C_{hier}(q) = C_{route}^{(\ell)}(q) + \Pr(\text{drill} \mid q) C_{drill}(q). \quad (7)$$

This creates a plausible efficiency benefit when abstractions are faithful and drill-down is relatively rare, but it is not guaranteed. Hierarchy can also introduce abstraction error, hide minority evidence, or perform poorly on cross-functional tasks that do not align with the organization tree. Recent systems such as OrgAgent and HORMA provide independent evidence that hierarchical organization can

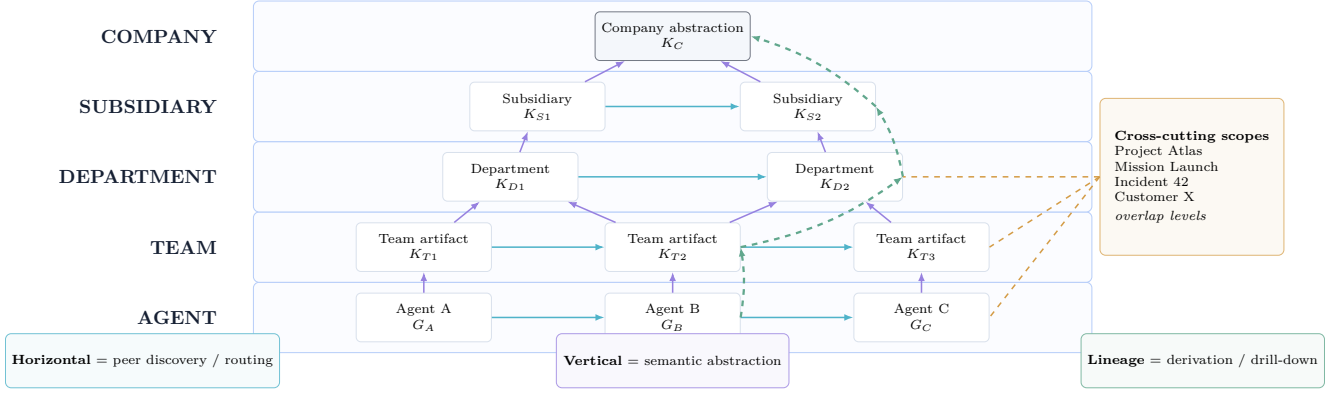


Figure 1: The 3D Lineage Fabric. Horizontal edges coordinate peers at the same abstraction level; vertical edges create higher-order organizational representations; lineage threads preserve derivation back to exact local evidence. Projects and missions remain cross-cutting scopes rather than being forced into the organization tree.

improve coordination or context efficiency in some settings, while leaving the Mycelic-specific question—persistent organizational knowledge with reversible lineage—open [21, 22].

3.2 Benefits that must be earned experimentally

The architecture is designed to test, not assume, five benefits:

1. **Local ownership:** raw evidence remains inside the owning NeuralGraph unless policy explicitly exports it.
2. **Representation scaling:** higher levels can answer broader queries from compact abstractions rather than repeatedly scanning every leaf memory.
3. **Reversible abstraction:** lineage allows a summary or conclusion to drill back to the exact lower-level support used to create it.
4. **Scope-aligned routing:** team/department/company queries can begin near the corresponding abstraction level, while project and mission hyperedges support cross-functional routes.
5. **Failure diagnosis:** lineage exposes which sources, agents, credentials, or transformations must survive for a higher-level capability to remain reconstructable.

The hierarchy should be removed or simplified if matched experiments show that flat federation reaches the same task-success/cost frontier.

4 Mycelic Architecture

Figure 2 is intentionally asymmetric: raw memories live at the leaves; the global layer stores only the information necessary to discover, route, relate, validate, and reconstruct knowledge across those leaves. JetStream may retain messages according to stream policy and replay them to consumers, but replayable transport state is not treated as the canonical semantic truth store [9].

4.1 NeuralGraph: local persistent memory

Each NeuralGraph owns the exact local objects needed for grounding: raw or normalized observations, memory nodes, entities, timestamps, causal links, local derivations, and authorization tags. Local retrieval may use vector similarity, graph traversal, temporal search, or other operators, but those choices remain behind a stable adapter.

4.2 JetStream: durable event transport

JetStream provides temporal decoupling, replay, durable consumers, acknowledgments, redelivery, and subject filtering. Those properties make it a suitable substrate for asynchronous knowledge propagation and disconnected recovery, while preserving the distinction between *message durability* and *knowledge validity* [9, 10].

4.3 Mycelic Fabric: global knowledge topology

The fabric is a sparse index over exported KnowledgeArtifacts and agent capabilities. It answers questions such as: which agents are semantically relevant, which scopes authorize access, which claims are temporally valid, and which candidate supports are lineage-independent? It is not required to store the underlying raw memories.

4.4 Tesseract: four-face router

Tesseract is deliberately narrow. Let

$$\mathcal{R}_4 = \{r_{1hop}, r_{multi}, r_{temp}, r_{open}\}. \quad (8)$$

Given query q , policy state π , fabric state F_t , and a routing cost model C , the router selects

$$r^*(q) = \arg \max_{r \in \mathcal{R}_4} \mathbb{E}[U(r, q, F_t, \pi)] - \mu C(r). \quad (9)$$

The four faces form a coarse retrieval-policy layer; the selected face determines how evidence is sought, while the fabric router determines which authorized agents should participate.

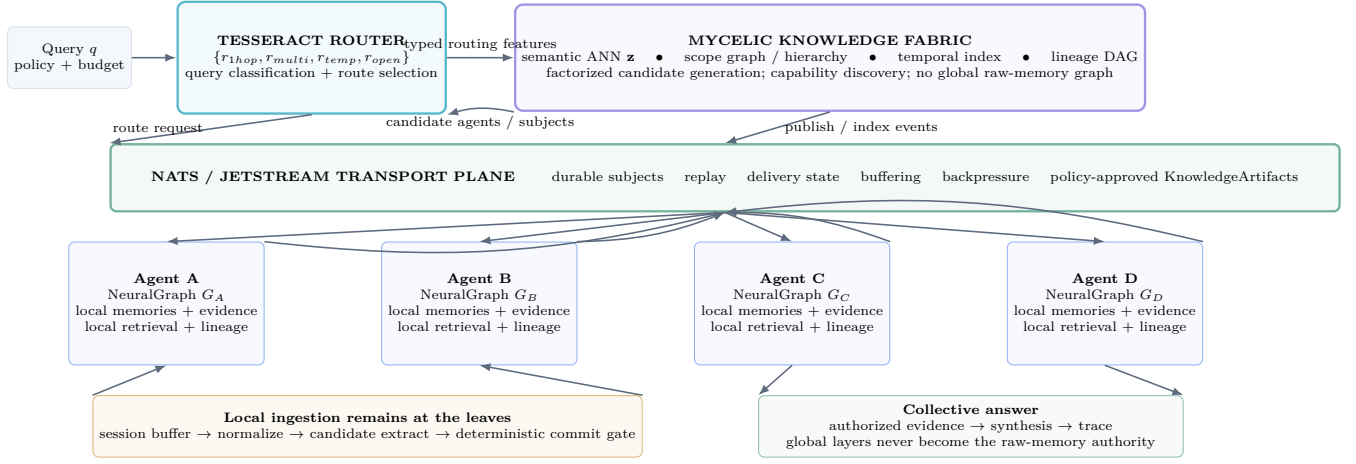


Figure 2: Mycelic system boundary. The global architecture maps and routes knowledge without merging private NeuralGraphs. JetStream is the durable communication substrate; Tesseract is only the four-face router.

5 Factorized Routing Topology

Mycelic models routing state with typed factors rather than forcing semantically different structures into one distance function. Semantic similarity is continuous and high-dimensional; organizational and project membership is overlapping and partly hierarchical; time is interval-ordered; lineage is a directed derivation structure; and policy and availability are hard constraints. For artifact a , define the typed routing state

$$\chi(a) = (\mathbf{z}_a, \Sigma_a, I_a, \Lambda_a), \quad (10)$$

where $\mathbf{z}_a \in \mathbb{R}^d$ is semantic state, Σ_a is an overlapping scope object, $I_a = [t_a^-, t_a^+)$ is temporal validity, and Λ_a is a lineage DAG or its lossless queryable representation. Policy, availability, and execution cost are modeled separately as constraints and decision features.

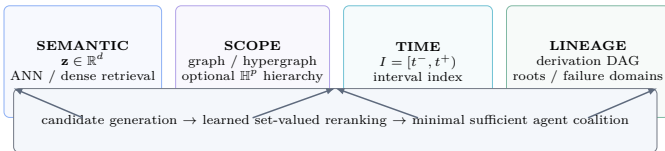


Figure 3: Typed routing topology. Each information family keeps its natural structure; the router fuses evidence only at decision time.

5.1 Why mixed geometry is preferable

Hierarchy is the one place where non-Euclidean geometry is especially plausible. Hyperbolic spaces expand exponentially with radius and can represent tree-like depth with less distortion than low-dimensional Euclidean coordinates. Recent retrieval systems report gains from hyperbolic or hybrid Euclidean–hyperbolic representations [11, 12]. This does *not* imply that the entire Mycelic Fabric should be hyperbolic. Instead, a hierarchy-aware embedding $\mathbf{h}_a \in \mathbb{H}^p$ may be learned for the hierarchical component of scope,

while cross-functional memberships (project, mission, incident, customer) remain explicit sets or hyperedges. The hypothesis is therefore testable by ablation rather than assumed by design.

5.2 Two-stage candidate generation and set-valued routing

For query q , inexpensive indexes first generate a candidate pool

$$\mathcal{C}(q) = \text{TopK}_{sem}(q) \cup \mathcal{C}_{scope}(q) \cup \mathcal{C}_{time}(q), \quad (11)$$

followed by authorization filtering $\mathcal{C}_\pi(q) \subseteq \mathcal{C}(q)$. Lineage is primarily a reranking/diversification signal: two semantically excellent candidates may add little value if both depend on the same root.

Routing is naturally *set-valued*: a distributed-only question may require several agents, while contacting too many agents wastes latency, tokens, bandwidth, and privacy budget. Let A be a selected agent set and B_q a route budget. We propose

$$\begin{aligned} \hat{A}(q) = \arg \max_{A \subseteq \mathcal{C}_\pi(q), |A| \leq B_q} & \sum_{i \in A} s_\theta(q, i) + \eta \text{Cov}(q, A) \quad (12) \\ & + \gamma \text{Div}_{lin}(A) - \mu \text{Cost}(A), \quad (13) \end{aligned}$$

where s_θ is a learned agent/evidence score; Cov rewards complementary capability coverage; Div_{lin} rewards lineage-disjoint support when corroboration matters; and Cost includes latency, messages, token use, and optional privacy exposure. Recent routing work independently supports two design choices relevant here: supervised/set-valued routing can outperform nearest-neighbor or zero-shot selection, and retrieve-then-rerank architectures can outperform metadata-only selection at scale [14, 15, 16]. Mycelic’s research question is narrower: whether these learned routing ideas gain additional value from persistent distributed knowledge state, explicit authorization, time, and lineage.

5.3 Tesseract’s four faces are paradigms, not coordinates

The four Tesseract faces remain useful as a coarse retrieval-policy choice:

$$\mathcal{R}_4 = \{r_{1hop}, r_{multi}, r_{temp}, r_{open}\}. \quad (14)$$

The face determines *how* evidence should be sought; Equation 13 determines *which authorized agents* should participate. This separation avoids asking one geometry to solve both query-type classification and distributed agent selection.

6 Lineage, Reconstruction, and Capability Survival

Let k be a task-relevant proposition and $\mathcal{P}_k$ the set of valid authorized derivation paths that can reconstruct it. Each path p depends on a set of failure domains $D(p)$: agents, sources, credentials, tools, databases, scopes, or shared ancestors.

A useful survivability margin is the minimum-cost set of failures that intersects every surviving derivation:

$$\kappa(k) = \min_{F \subseteq \mathcal{F}} \{c(F) : \forall p \in \mathcal{P}_k, D(p) \cap F \neq \emptyset\}. \quad (15)$$

Equation 15 is intentionally recognizable as a hitting-set/min-cut style object; the novelty cannot be “we invented min-cut for memory.” Its value is as a rigorous baseline for measuring whether semantic transformation, authorization, decoder state, and dynamic routing explain residual capability failure beyond classical structure.

6.1 Independent support versus copied support

Ten memories descending from one upstream source can have replica count ten but epistemic support one. For two derivations p_i, p_j , strict independence with respect to a selected domain family $\mathcal{F}^*$ is

$$D_{\mathcal{F}^*}(p_i) \cap D_{\mathcal{F}^*}(p_j) = \emptyset. \quad (16)$$

In practice independence is domain-relative: two paths may use different agents but the same database, or different documents written from the same human source. Mycelic therefore records typed derivation edges rather than flattening provenance into an undifferentiated list of source IDs.

6.2 Active lineage diversification

A proposed self-healing controller detects high-value claims with low $\kappa(k)$ and requests a blind verification through a disjoint source/tool/agent path. Let action a create a candidate new derivation at cost $C(a)$. A simple policy is

$$a^* = \arg \max_a \frac{\mathbb{E}[\kappa'(k) \mid a]}{C(a)}. \quad (17)$$

The controller should not reveal the existing answer to the verifier; otherwise it creates correlated confirmation rather than independent support. Disagreement becomes an explicit conflict object rather than an automatic merge.

7 Benchmark and Results Plan

The evaluation is designed as a **fillable architecture leaderboard**. Literature-reported numbers are not inserted into the headline board because models, datasets, retrieval budgets, privacy boundaries, and failure regimes differ across papers. Every architecture below will be reimplemented or emulated under the *same* evidence, model, policy, query budget, and failure schedule. The paper will then populate each score cell from measured runs.

7.1 Architecture families under comparison

The experiment compares complete architectural classes, not only router variants:

1. **Centralized raw memory (C-RAW)**: all persistent memories are copied into one shared store/index queried by every agent.
2. **Centralized graph memory (C-GRAPH)**: one global graph stores entities, relations, time, and provenance for all agents.
3. **Replicated central memory (R-CENTRAL)**: the same global state is replicated for infrastructure resilience; semantic authority remains centralized.
4. **Federated query-time retrieval (FED)**: memories remain local; a broker fans a query out to stores and combines returned evidence, with no persistent organizational abstraction layer.
5. **Flat distributed NeuralGraphs (FLAT)**: local NeuralGraphs exchange approved artifacts through JetStream and peer routing, but there is no vertical organizational abstraction.
6. **Hierarchical abstraction without lineage drill-down (HIER)**: user/team/department/company representations exist, but higher-level artifacts cannot deterministically reconstruct exact lower-level derivations.
7. **Hierarchical lineage fabric with fixed routing (MYC-FIXED)**: the 3D Lineage Fabric is active, but routing uses fixed factor weights.
8. **Full Mycelic (MYC-LEARNED)**: the 3D Lineage Fabric plus factorized learned set-valued routing and optional active lineage diversification.

7.2 Headline architecture scorecard

For scanability we will report a secondary 0–100 *Balanced Systems Score* after normalizing compatible metrics to

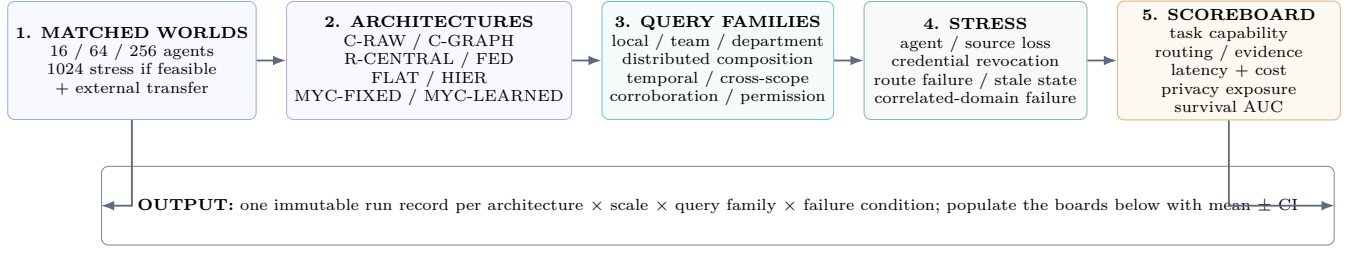


Figure 4: Matched architecture benchmark. The experiment changes the memory/coordination architecture while holding models, information, policy, and budgets fixed.

Architecture	Task Acc.	Evidence / Route	p95 ms	Cost / q	Raw Exposure	Failure	AUC	Balanced Score / Verdict
Centralized raw memory								
Centralized graph memory								
Replicated central memory								
Federated query-time retrieval								
Flat distributed Neural-Graphs								
Hierarchy, no lineage drill-down								
Mycelic hierarchy + fixed router								
Full Mycelic								
Oracle evidence / coalition			–	–	–	–	–	upper bound

Table 1: Primary architecture leaderboard. Every score is generated under the matched protocol; no cross-paper accuracy number is treated as directly comparable. “Evidence / Route” is Recall@k for centralized conditions and sufficient-coalition F1/recall for distributed conditions.

[0, 1]:

$$S_{bal} = 100 (A_{task} R_{evid} E_{cost} P_{priv} U_{surv})^{1/5}, \quad (18)$$

where E_{cost} increases as normalized latency/compute/communication cost decreases, P_{priv} increases as raw-memory exposure decreases, and U_{surv} is normalized survival AUC. The geometric mean intentionally penalizes an architecture that is excellent on one dimension but catastrophically weak on another. **Primary conclusions will still be based on the full Pareto frontier and confidence intervals, not the composite alone.**

7.3 Organizational hierarchy scorecard

This board directly tests whether the 3D Lineage Fabric earns its complexity. Organization-aligned questions are separated from cross-functional project/mission questions so a hierarchy cannot win merely because the benchmark

mirrors the tree.

7.4 Scenario coverage matrix

7.5 Metric definitions and protocol

For centralized conditions report evidence Recall@k, MRR/nDCG where appropriate, end-task accuracy, calibration, latency, and cost. For distributed conditions with gold valid coalition family $\mathcal{A}_q^*$ and selected set $\hat{\mathcal{A}}_q$, additionally report set precision, recall, F1, Jaccard, exact match, sufficient-coalition recall, over-routing ratio $|\hat{\mathcal{A}}_q|/|\mathcal{A}_q^{min}|$, and regret relative to a minimum-cost oracle [14]. Systems measurements include p50/p95 latency, agents contacted, messages, bytes transferred, model tokens, raw-memory exposure, and compute cost. Authorization violations must remain zero in deterministic tests.

Hierarchy experiments add two integrity measures. **Lineage drill-down accuracy** is the fraction of higher-level

Condition	Org Acc.	Cross-scope Acc.	Agents	Bytes/q	Drill-down	Faithful.	Verdict
Flat federation / no hierarchy	☐	☐	☐	☐	–	–	☐
Hierarchy, no lineage	☐	☐	☐	☐	☐	☐	☐
Hierarchy + lineage / fixed routing	☐	☐	☐	☐	☐	☐	☐
Full Mycelic	☐	☐	☐	☐	☐	☐	☐

Table 2: *Hierarchy-specific board. Drill-down measures whether a higher-level artifact can recover the exact supporting lower-level artifact/memory IDs; faithfulness measures whether the abstraction preserves the task-relevant proposition without introducing unsupported content.*

Query family	16	64	256	External	Primary question	Headline
Local / direct	✓	✓	✓	✓	Can the architecture avoid unnecessary fan-out when one holder is sufficient?	☐
Team / department / company	✓	✓	✓	–	Does semantic abstraction improve broad organizational reasoning at lower global-state cost?	☐
Distributed composition	✓	✓	✓	✓	Does the selected evidence contain a valid sufficient coalition?	☐
Temporal / supersession	✓	✓	✓	✓	Does valid-time structure prevent stale but semantically attractive evidence?	☐
Cross-project / mission	✓	✓	✓	✓	Can overlapping scopes recover information that a strict organization tree would miss?	☐
Independent corroboration	✓	✓	✓	–	Does lineage-aware selection prefer genuinely independent support?	☐
Permission-conditioned	✓	✓	✓	–	Can the system recover a valid coalition when the nearest holder is unauthorized?	☐
Stale / conflicting	✓	✓	✓	✓	Can the system reject or downweight contradictory and superseded support?	☐
Failure recovery	✓	✓	✓	–	How much capability survives controlled agent, source, route, and credential loss?	☐

Table 3: *Planned scenario coverage. Blank headline cells are reserved for the measured result that best summarizes each query family.*

answers for which the system deterministically recovers the exact gold supporting artifact/memory IDs. **Abstraction faithfulness** measures whether the higher-level semantic artifact preserves the task-relevant proposition while avoiding unsupported additions; this should be judged with structured gold facts where possible and a versioned semantic evaluator otherwise.

Controlled worlds use frozen train/validation/test seeds and held-out organizations/scopes. Hyperparameters are selected only on validation worlds. External transfer partitions LongMemEval, LongMemEval-V2, and LoCoMo/LoCoMo-Plus histories across independently owned agents while preserving the original task answer [17, 18, 19, 20]. Report bootstrap confidence intervals and paired query-level significance tests.

7.6 Ablation board

Remove / alter	Δ task acc.	Δ cost
Vertical abstraction hierarchy	☐	☐
Lineage drill-down	☐	☐
Cross-cutting scope hyperedges	☐	☐
Temporal validity	☐	☐
Lineage diversity in routing	☐	☐
Hierarchy-aware scope representation	☐	☐
Learned reranking (fixed fusion only)	☐	☐
Set-valued selection (single-agent only)	☐	☐

Table 4: *Ablation board. Every structural mechanism must earn its complexity empirically.*

Publication decision rule. Mycelic is scientifically interesting only if it moves a held-out capability/cost/privacy/resilience frontier beyond strong centralized, federated, and flat-distributed alternatives. The hierarchy claim fails if flat federation matches organizational task quality and cost; the lineage claim fails if drill-down, failure prediction, or independent-support routing adds no measurable value; the learned-routing claim fails if fixed or semantic-only routing matches it under equal budgets.

8 Research Hypotheses

The architecture supports four increasingly difficult hypotheses.

H1 - hierarchical abstraction improves the organizational capability/cost frontier. Reversible semantic abstraction across agent, team, department, subsidiary, and company levels reduces global state scanned, agents contacted, bytes, or tokens for organization-level queries without degrading task accuracy, while cross-cutting project/mission scopes prevent unacceptable losses on non-tree-aligned tasks.

H2 - factorized learned routing improves coalition selection. A two-stage set-valued router that fuses semantic, scope, time, lineage, policy, and cost features improves the task-success/cost frontier over semantic-only, metadata-filtered, fixed-factor, and graph-policy baselines, with the largest gains on distributed, temporal, cross-functional, and corroboration queries.

H3 - reconstructability predicts survival. A task-conditioned predictor derived from actual execution traces and authorized derivations predicts held-out capability survival better than replica count, source count, reachability, and classical correlated-root baselines.

H4 - active diversification improves resilience. Under matched verification/storage cost, proactively acquiring a genuinely disjoint reconstruction path preserves more capability under correlated failures than replicating an existing derivation.

H1 is the architectural hierarchy claim. H2 is the routing claim. H3 is the stronger scientific reliability claim. H4 is the most distinctive intervention if it survives cost-controlled experiments.

9 Implementation Requirements

The publication path requires one canonical NeuralGraph service for ingestion, retrieval, persistence, lineage, and traces; Tesseract accesses that service through a stable adapter, and benchmark code calls the same API. This prevents evaluation/runtime divergence and makes every reported result attributable to a single executable path.

Two invariants are non-negotiable. First, in-memory and SQLite persistence must preserve identical provenance semantics. Second, consolidation must not replace deriva-

tion structure with a set-union of source IDs; copied-from, summarized-from, inferred-from, abstracted-from, verified-by, and co-observed-with relations imply different dependence.

The minimal publication-grade substrate is therefore: stable IDs, differential storage tests, typed derivation events, reversible failure views, machine-readable traces tied to evidence actually consumed, full export-envelope validation, and deterministic research mode with decay/consolidation frozen unless explicitly under study.

10 Related Work and Novelty Boundary

Graph-based and reconstructed memory are already active areas, so Mycelic should not claim graph memory, persistent memory, or reconstruction as standalone novelty [3, 4]. Governed shared memory formalizes scope, temporal supersession, provenance, and policy propagation in multi-agent systems [5]; MemLineage makes derivation lineage operational for memory security [6]; and MAP-Graph uses provenance-aware eligibility, trust, and risk-sensitive gating in shared multi-agent workflows [7]. Classical lineage-driven fault injection already reasons backward from outcomes to causally relevant failures [8].

Hierarchy is likewise prior art rather than a novelty claim. OrgAgent reports that company-style organization can improve multi-agent coordination and reduce token use in some reasoning settings [21]; HORMA organizes agent experience hierarchically and navigates toward minimal sufficient context [22]; TaskWeave studies long-horizon organizational coherence with dependency-aware trace memory [23]. Mycelic therefore does not claim that “hierarchy works.” The testable contribution is the combination of persistent local ownership, semantic abstraction across organizational levels, cross-cutting scopes, and exact lineage drill-down under matched centralized/federated baselines.

Routing itself is also crowded. RAGRouter-Bench shows that retrieval strategy should be selected adaptively across query-corpus contexts [13]; SkillRouter demonstrates the strength of retrieve-then-rerank selection at large skill catalogs [15]; set-valued multi-agent routing work shows that supervised routing can outperform nearest-neighbor and zero-shot selection when multiple agents may be required [14]; and GraphPlanner learns adaptive multi-agent workflows from heterogeneous graph memory [16]. HyperbolicRAG and HypRAG further motivate testing non-Euclidean representations for hierarchical retrieval without assuming Euclidean geometry is always sufficient [11, 12].

The defensible opportunity is therefore the **combination and phenomenon**: private non-converging local memories; a sparse global knowledge topology rather than a raw global graph; set-valued routing over persistent semantic, scope, temporal, authorization, and lineage state; and direct measurement of whether that structure improves both routing efficiency and distributed-only capability survival under correlated semantic and authorization failures. The

paper must show a new empirical frontier, not rename known routing or distributed-systems concepts.

11 Limitations

The factorized topology is not a claim that one universal geometry exists; its value depends on measurable routing or prediction gains. Hyperbolic scope embeddings are optional and must earn their complexity experimentally. Scope structures may be non-tree-like and organization-specific. Lineage independence is only as accurate as the failure-domain taxonomy and derivation instrumentation. Active diversification can increase cost and may generate contradictory evidence. Finally, a distributed architecture can reduce raw-memory centralization while still creating sensitive global metadata; access-control and metadata minimization remain first-class requirements.

12 Conclusion

Mycelic reframes persistent organizational agent memory as a multilayer knowledge problem. NeuralGraph stores local evidence-bearing memories. JetStream moves durable events. The 3D Lineage Fabric lets peers coordinate horizontally, lets knowledge become semantically more abstract vertically, and preserves derivation threads back to locally owned evidence. Projects and missions cut across the organization tree as overlapping scopes. Tesseract chooses a retrieval face, then a factorized learned router selects the authorized agent coalition. The collective’s knowledge is therefore not a single database: it is the set of claims and abstractions that remain reconstructable from authorized distributed evidence.

The architecture earns scientific relevance only if this structure predicts or improves capability survival beyond ordinary federation, replication, and classical reliability metrics. That is the experiment the system should now be built to pass—or to fail cleanly.

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Research-status note. This manuscript is an architecture and experimental-design preprint. It does not fabricate benchmark results. H1–H4 are falsifiable hypotheses, and the paper should be revised around the evidence produced by the controlled experiments described above.